SmolVLM

Redefining Small and Efficient Multimodal Models

Hugging Face & Stanford University

Compact vision-language models achieving competitive performance with dramatically lower memory footprints through strategic architectural optimizations.

Large VLMs Are Powerful but Impractical for Edge Deployment

- State-of-the-art VLMs require massive compute and memory
- Deployment on mobile/edge devices is severely limited
- Existing efficient models still demand multi-GB RAM
- On-device inference with no cloud dependency remains a challenge

Idefics-80B

80B parameters — 300× larger than SmolVLM-256M

Molmo-A1B-7B

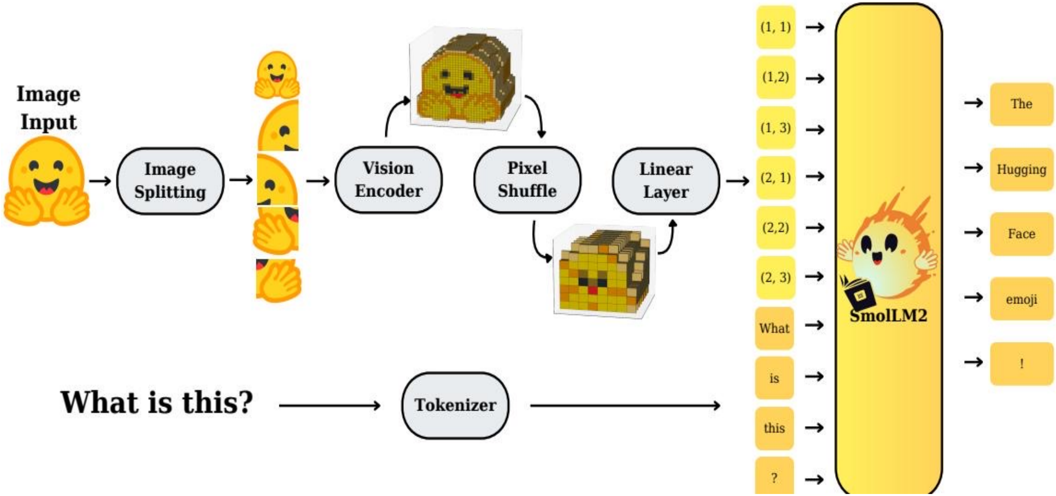
Requires 27.7 GB GPU RAM at batch size 1

SmolVLM-256M

0.8 GB GPU RAM — outperforms Idefics-80B

Source: arXiv:2504.05299

SmolVLM Architecture: Image Splitting, Pixel Shuffle, and SmolLM2 Backbone



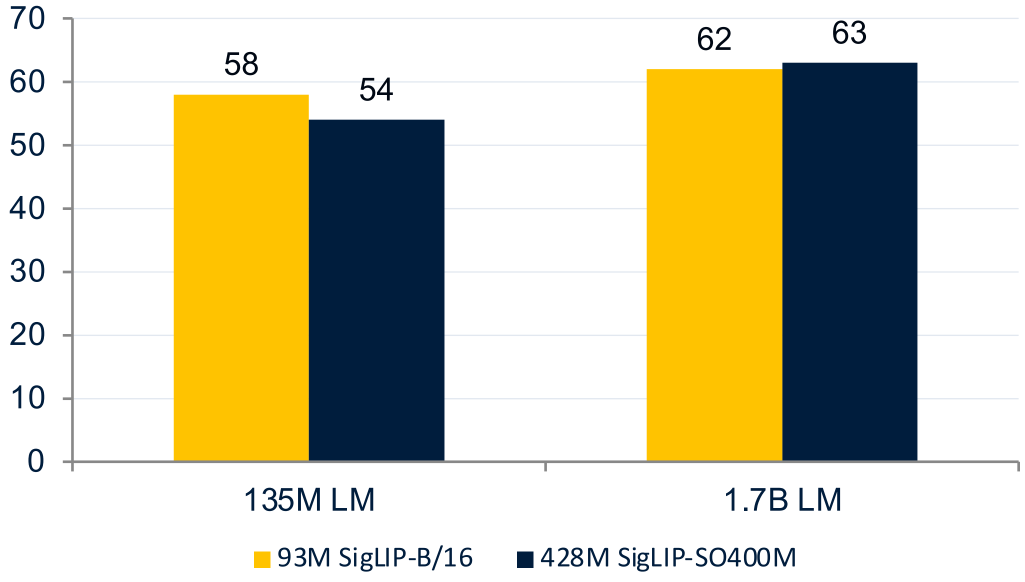
1. **Image Splitting**
   Images split into sub-images for processing
2. **SigLIP Encoder**
   Vision features extracted by SigLIP encoder
3. **Pixel Shuffle**
   Spatial features → channels (token compression)
4. **Linear Projection**
   Projected into SmolLM2 input space

Video: frames are sampled and processed through the same pipeline.

Source: arXiv:2504.05299

Smaller Vision Encoders Outperform Larger Ones in Compact VLMs

Finding 1 — Balanced Encoder-LM Compute



Source: arXiv:2504.05299

93M SigLIP-B/16 + 135M LM

Outperforms the 428M SigLIP-SO400M with the same 135M LM. Smaller encoder is better for compact VLMs.

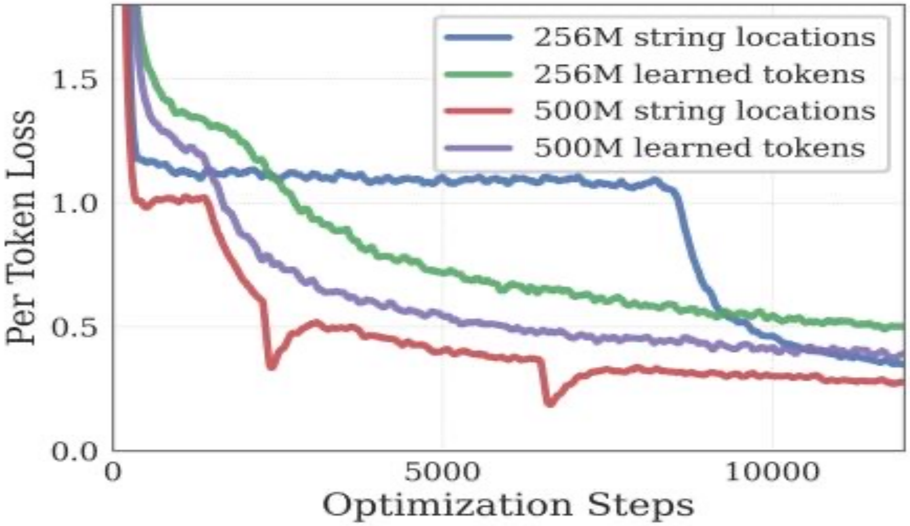
428M SigLIP-SO400M + 1.7B LM

Only at 1.7B LM scale does the larger encoder justify its 10% parameter overhead.

4.6× parameter reduction (428M → 93M) with better performance

Aggressive Pixel Shuffle (r=4) Cuts Visual Tokens 16× for Small Models

Finding 3 — Aggressive Token Compression



Pixel Shuffle Mechanism

- Rearranges spatial features into channels
- Reduces visual tokens by r² (shuffle factor squared)
- r=2 → 4× token reduction (used by larger models)
- r=4 → 16× token reduction (optimal for

Impact

16× fewer tokens eases attention overhead and improves long-context modeling for compact models.

Source: arXiv:2504.05299

Extending Context from 2K to 16K Tokens Unlocks Higher Image Resolution

Finding 2 — Extended Context Length

2K tokens	8K tokens	16K tokens
Baseline	Smaller variants	2.2B model

RoPE Base Adjustment

RoPE base increased from 10K to 273K to support longer sequences without positional degradation.

Mixed Context Fine-Tuning

Fine-tuned on mixed long/short-context data to maintain quality across sequence lengths.

Performance improves significantly with extended context, enabling higher-resolution image understanding and longer video sequences.

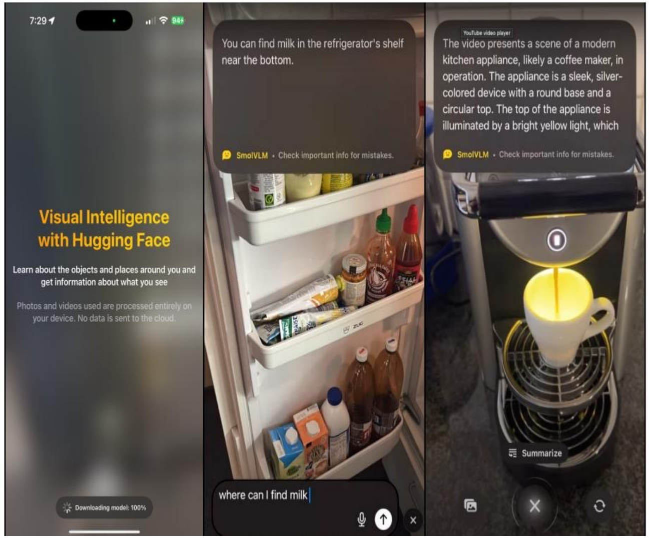
Source: arXiv:2504.05299

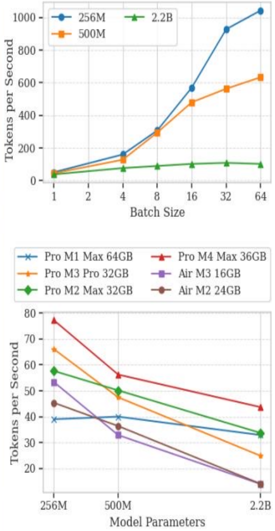
SmolVLM Achieves Competitive Performance at Sub-1GB to 4.9GB RAM

Capability	Benchmark	SmolVLM 256M	SmolVLM 500M	SmolVLM 2.2B	Best Efficient OS
Single-Image	OCRBench	52.6%	61.0%	72.9%	54.7% (Molmo)
Single-Image	AI2D	46.4%	59.2%	70.0%	71.0% (Molmo)
Single-Image	ChartQA	55.6%	62.8%	68.7%	48.0% (Molmo)
Single-Image	TextVQA	50.2%	60.2%	73.0%	61.5% (Molmo)
Single-Image	DocVQA	58.3%	70.5%	80.0%	77.7% (Molmo)
Single-Image	ScienceQA	73.8%	80.0%	89.6%	87.5% (Molmo)
Multi-task	MMMU	29.0%	33.7%	42.0%	33.9% (Molmo)
Multi-task	MathVista	35.9%	40.1%	51.5%	37.6% (Molmo)
Multi-task	MMStar	34.6%	38.3%	46.0%	43.1% (Molmo)
Video	Video-MME	33.7%	42.2%	52.1%	45.0% (InternVL2)
Video	MLVU	40.6%	47.3%	55.2%	48.2% (InternVL2)
Average	**All Benchmarks**	**44.0%**	**51.0%**	**59.8%**	—

GPU RAM Usage (batch=1)

Real-Time Inference on iPhones: Up to 80 Tokens/Second on Apple Silicon





HuggingSnap iOS App

Processes photos and videos entirely locally with no cloud dependency.

~80 **tokens/sec**
Pro M1 Max 64GB

~1,000 **tokens/sec**
Batch size 64, high-end

~55 **tokens/sec**

Throughput scales roughly linearly with batch size for the 256M model.

Source: arXiv:2504.05299

Key Takeaways

Design Principles for Efficient Multimodal Models

1. Balance encoder-LM compute: smaller vision encoders outperform larger ones in compact VLMs.
2. Extend context length: RoPE scaling from 2K to 16K unlocks higher-resolution understanding.
3. Use aggressive token compression: r=4 pixel shuffle (16× reduction) is optimal for small models.
4. Sub-1GB inference is achievable: SmolVLM-256M uses only 0.8 GB RAM and surpasses Idefics-80B.
5. Fully open-source: all models, data, code, and the HuggingSnap mobile app are publicly available.

Implication: Architectural efficiency matters more than raw scale for on-device multimodal AI.